

EDUCATION

- **Indian Institute of Information Technology, Lucknow** Lucknow
Bachelor of Information Technology; CGPA: 8.32 Apr 2020 - Apr 2024
- **Relevant Courses:** Data Structures & Algorithms, Operating Systems, System Design, DBMS, Object-Oriented Programming, Distributed Systems.

PROFESSIONAL SUMMARY

- Backend engineer with **4+ years** building scalable distributed systems, microservices, and APIs in production.
- Delivered measurable reliability and performance gains, including a **208× query speedup** on a 524M-row table and zero-downtime database migrations across **4 production regions**.
- Hands-on with **Java/Spring Boot, Ruby on Rails, SQL, Redis, ClickHouse, AWS (RDS), Jenkins, and CI/CD**, with strong ownership of deployments and operational excellence.

EXPERIENCE

- **BrowserStack** Mumbai
SDE - Backend - Ruby, Rails, Express, JavaScript, Spring Boot, Java, ClickHouse, Helm, Jenkins Apr 2025 - Present
 - Led end-to-end bring-up of a new **Australia (GRR-AUS) region** for TestHub over ~3 months — regional infra (RDS, Redis, IAM, quotas), Kafka topics, ClickHouse pipeline, ingestor routing, cross-region peering, and monitoring — now serving **360K requests/hour** in production.
 - Cut customer dashboard load times from **3–4 min → 3 s** by replacing full table scans with index scans on the TestHub DB, resolving a deal-blocking performance issue.
 - Drove a **208× query speedup** (9,652ms→46ms) on a **524M-row** table via a zero-downtime **gh-ost** migration across **4 regions**.
 - Averted a projected outage via zero-downtime **INT→BIGINT** migrations on **3 billion-row** tables (~1TB) growing **2.89M rows/day**.
 - Shipped **project-name sanitization** across project create/search/list endpoints serving **100M+ API calls**, enabling **multi-language support across all products** and unifying dashboard/SDK validation so user builds stay in sync.
 - Cut **RDS CPU from 96%** by offloading aggregate queries to **ClickHouse** and moving the Query Killer to worker-based execution.
 - Built **connectors-as-code (IaC)** for **35 Debezium CDC connectors** across **4 regions** (USE/APS/EUW/APSE2) with auto-sync after infra merges, eliminating manual config drift; split Kafka history topics across 3 regions for pipeline scalability.
 - Halved auth Rails calls (**2 → 1**) for A11y/AppA11y by rerouting SDK auth through Config Server → Ingestor, eliminating **102,537 redundant auth calls/day** and reducing API/ingestor latency.
 - Remediated critical **SQL injection** across **50+ API endpoints** spanning 4 teams on a system handling **4,000 req/s**; drove org-wide **Semgrep SAST** for CI-level SQLi prevention covering **56 P0/P1 API actions (~98% of traffic)**.
 - Resolved **2 P0 vulnerabilities** (OS Command Injection/RCE and Axios SSRF bypass) before exploitation; enabled **TLS for DB connections** and patched multiple high-severity CVEs (node-tar, Handlebars, minimatch ReDoS).
- **Zus (formerly 0chain)** Cupertino, CA (Remote)
Software Engineer - Blockchain Storage & Mobile - Golang, Kotlin, GitHub Actions, Gin, Grafana Feb 2022 - Mar 2025
 - Shipped 2 production Android apps, **Bolt** (crypto wallet) and **Vult** (storage), driving **20%** adoption growth and **\$2M+** in transactions.
 - Built a **UAT automation suite** for blockchain storage, cutting bugs by **90%** and release cycles from **2 weeks to 3 days**.
 - Optimized **S3-compatible storage** handling **500M+ files** and **40TB+** across **50+ nodes** at sub-100ms latency, **99.9%** uptime.
 - Automated dynamic **Sharder/Miner** management and view-change protocols, boosting developer productivity by **90%**.
 - Enhanced **multi-wallet concurrency** via unified **GoSDK**, supporting **10K+ concurrent** transactions and **300%** load-test gains.
- **Oppia Foundation (incl. GSoC) [Merged PRs]** Mountain View, CA (Remote)
OSS Contributor & GSoC - Kotlin, GitHub Actions, JUnit, Gradle, Bazel, Android SDK Jun 2021 - Sep 2022
 - Contributed to **Gradle to Bazel** migration, achieving **10x** improvement in build speeds.
 - Built **CI/CD pipelines** (GSoC) automating build triggers and deployment to **Google Drive, GCP Bucket, and Play Store** tracks.
 - Developed modular **MVVM/MVP** architectures with **tablet support** for multiple screen sizes.
 - Authored **70+ Unit Tests** and **20+ E2E tests** using Espresso, Robolectric, and UIAutomator.
- **GNOME Foundation** Remote
Open Source Contributor - Tech Stack: C, XML, GTK+ Sep 2021 - Feb 2022
 - Enhanced **Nautilus file manager** with new path-bar features, improving user efficiency by **20%**.
 - Implemented **drag-and-drop support** and keyboard shortcuts for seamless file organization.

PROJECTS

- **Zus HelloWorld (Android):** Reference **Kotlin** app demonstrating **Zus/0chain GoSDK** integration — **Bolt** crypto wallet (ZCN/Ethereum staking) and **Vult** decentralized storage with multi-file upload, download, and thumbnail generation.

ACHIEVEMENTS

- Selected for **Google Summer of Code 2022** (Oppia Foundation).
- Won **top prize (16K INR)** at the **CAMS Hackathon** for developing its Android app.
- Rated **Specialist** on [Codeforces](#) (max-rating **1509**) and **5-star** on [CodeChef](#).
- Practised **700+** DSA problems on [Leetcode](#).

TECHNICAL SKILLS

- **Programming & Scripting:** Java, Go, Kotlin, Python, JavaScript, SQL, Bash, C/C++
- **Backend Frameworks:** Spring Boot, Ruby on Rails, Gin, Express.js, REST APIs, GraphQL, Microservices
- **DevOps & Cloud:** AWS (RDS), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, Helm, Prometheus/Grafana/Loki
- **Databases & Testing:** MySQL, PostgreSQL, MongoDB, Redis, ClickHouse, Query Optimization, JUnit, Mockito
- **Tools & Libraries:** Git, Apache Kafka, Debezium, CDC, gh-ost, Semgrep, Gradle, Bazel, System Design